EDR Level 1 Group Project

2012 Toyota Yaris crash

A 2012 Toyota Yaris owner called his insurance company to report that his vehicle was parked in front of his house, and when he woke up he found the vehicle front end to be damaged. The insurance company was suspicious and sent an investigator out to check, the investigator found the driver airbag deployed and believes airbags will only deploy with the key on. He asked the owner if he had possession and control of the keys at all times, and the driver swears that he has the only keys and they were in his possession the entire time. The insurance company suspect fraud and has hired you to evaluate the EDR report.

Recall that Toyota is different than other companies, that it records X Delta V in frontal events and Y Delta V in side events, and that if an angular crash occurs, it is likely to have BOTH a side event AND a frontal event from the same crash event. You do NOT have key cycles to help determine if this recording is from your event, you must match visible vehicle damage and the EDR up. Determine if the none, one, two, three, or all four came from your event. On the photographs, be prepared to point to which damage came from which events.

For the largest frontal damage, draw a sketch with the damage centroid identified. Using the Y and X Delta V, determine the PDOF (Hint: The Y data only goes to approximately 70 ms after wakeup, be sure to take the X at or near the same time). Draw the PDOF line on the sketch of the vehicle and indicate the total Delta V for the crash. Since the Y Delta V does not go to the end of the crash, use the formula Total DV = DVx/(cos PDOF).

For the lesser damage, what is the relationship between X and Y Delta V (can you do a PDOF or not?). Sketch the PDOF line through the damage centroid. Is there likely to be significant rotation as a result of this crash?

Are there any anomalies in the EDR either explained or unexplained in the data limitations? If so highlight them in your presentation (hint: Max speed)

For the biggest impact, Using the formula Speed V1 = Speed V3-DeltaVx, where the EDR has V1 and Delta Vx, to get V3 after impact? After considering the post impact speed and your PDOF sketch, did the vehicle hit something and come to a stop, or hit something and glance off but kept going, or hit something and break it off and keep going, or none of the above?

Convert the last reported speed from MPH to feet per second. Multiply the speed in FPS times the time between events. About how far apart are the two objects that were struck?

Given the time between events and the speeds, do you suspect this event took place on a limited access highway, on a country road, or in an urban environment? So far the crash location is unknown.

Last revised 4-30-2017

Practical Exercise

Each group, through their spokesman, should put together a short presentation (20 minutes) on your assigned vehicle. You should cut to the chase, and answer the following questions in a round of high-speed EDR file analysis. You can make reference to any material in the folder your have been assigned.

Answer the following questions for your vehicle

1. Show the photos of the vehicle damage to the rest of the class.
2. Sketch the vehicle on the board indicating the damage area(s).
3. Recall the "Big Four" questions that we answer in all cases to assure that the data is from the crash of interest. The next few questions will help you fill out that table

THE BIG FOUR QUESTIONS	Event 1	Event 2	Event 3
Is the recording Complete? (Y/N or Not Available) (For the event Number think - D, ND, DLE, F/S/R, Roll etc) (Y/N or Not Available)			
Do we have an explanation for the Key Cycle difference between Key Cycles at event and Key Cycles at investigation			
Is the reported Delta-V similar to our crash magnitude? (estimate based on the size of the dent). (Y/N or Not Available)			
Does the pre-crash speed data match our crash story? (Y/N or Not Available)			

4. How many events? (ND, D, etc- how do you know? page and line)
5. Are each of those events completely written? (was there enough power? How do you know? page and line)
6. Are any of your recorded events from prior collisions? How do you know? If you have prior events you can ignore them for the remainder of the exercise.
7. Do each of your events have Crash-Pulse/Delta-V data?
    - Did your EDR file capture the entire crash? Was there enough duration (time)?
    - What is the Maximum Longitudinal Delta V? (Did you get it from the Graph or from the System Status Table?)
    - What is the Maximum Lateral Delta V (Did you get it from the Graph or from the System Status Table?)
8. Return to your sketch and identify the dent associated with the largest event and sketch the PDOF.
9. What is the Total Delta V? (page and line or you may need a calculator)
10. Characterize the total Delta V data as small, moderate, or large
11. Do you think the Delta-V magnitude matches the crash damage (is the dent the right size)?
12. Is your crash a central collision or an eccentric collision with rotation?

13. Do any or all of your events have Pre-Crash Data?
14. If you have Pre-Crash data, identify the largest collision by DeltaV and answer the following questions about that Pre- Crash data
    - Does your pre crash data match your DeltaV? (Collision with another car, collision with a fixed object, etc?
    - What was your driver doing at time of the first Pre-Crash data samples (accelerating decelerating, steady state?)
    - If there was “steady state” pre crash speed calculate the speed range associated with that steady state condition
    - Did your driver see the crash coming (did he react – how do your know)?
    - Was there maximum braking just before the crash? (how do you know?)
15. Calculate the range of speed for “Speed at Impact”
16. Do you have a “working theory of how this crash occurred (Single vehicle run off road, collision with fixed object, inline vehicle-to-vehicle collision, two moving vehicle collision? If so make a sketch on the board?
17. How confident are you in your working theory?


CHEVROLET
30871513x

BRIDGESTONE
DOT EDCA 3KRR3811
RADIAL TUBE


Car Freshner
Black Ice



SEATING CAPACITY : TOTAL 5
FRONT 2 : REAR 3

The combined weight of occupants and cargo should never exceed 383 kg or 845 lbs.

NOMBRE DE PLACES : TOTAL 5
AVANT 2 : ARRIÈRE 3

Le poids total des occupants et du chargement ne doit jamais dépasser 383 kg ou 845 lb.

SEE OWNER'S MANUAL FOR ADDITIONAL INFORMATION.

TIRE	SIZE	COLD TIRE PRESSURE
FRONT	P185/60R15	220kPa, 32PSI
REAR	P185/60R15	220kPa, 32PSI
SPARE	T125/70D15	420kPa, 60PSI

PNEU	DIMENSIONS	PRESSION DES PNEUS À FROID
AVANT	P185/60R15	220kPa, 32PSI
ARRIÈRE	P185/60R15	220kPa, 32PSI
DE SECOURS	T125/70D15	420kPa, 60PSI

IMPORTANT NOTICE: Robert Bosch LLC and the manufacturers whose vehicles are accessible using the CDR System urge end users to use the latest production release of the Crash Data Retrieval system software when viewing, printing or exporting any retrieved data from within the CDR program. Using the latest version of the CDR software is the best way to ensure that retrieved data has been translated using the most current information provided by the manufacturers of the vehicles supported by this product.

CDR File Information

User Entered VIN	JTDBT4K32C1******
User	HARRIS
Case Number	WRECKER YARD 3
EDR Data Imaging Date	12/04/2013
Crash Date	12/04/2013
Filename	2012 YARIS 06 EDR FRONTAL.CDRX
Saved on	Wednesday, December 4 2013 at 15:42:40
Collected with CDR version	Crash Data Retrieval Tool 12.1
Reported with CDR version	Crash Data Retrieval Tool 14.1
EDR Device Type	Airbag Control Module
Event(s) recovered	Front/Rear (2), Side (2)

Comments

R185/60R15
BACK POWER
DLC

Data Limitations

CDR Record Information:

- Due to limitations of the data recorded by the airbag ECU, such as the resolution, data range, sampling interval, time period of the recording, and the items recorded, the information provided by this data may not be sufficient to capture the entire crash.
- Pre-Crash data is recorded in discrete intervals. Due to different refresh rates within the vehicle's electronics, the data recorded may not be synchronous to each other.
- Airbag ECU data should be used in conjunction with other physical evidence obtained from the vehicle and the surrounding circumstances.
- If the airbags did not deploy or the pretensioners did not operate during an event that meets a specified recording threshold, it is called a Non-Deployment Event. Data from a Non-Deployment Event can be overwritten by a succeeding event that meets the specified recording threshold. If the airbag(s) deploy or the pretensioners are operated, it is called a Deployment Event. Deployment Event data cannot be overwritten or deleted by the airbag ECU following that event.
- If power supply to the airbag ECU is lost during an event, all or part of the data may not be recorded.
- "Diagnostic Trouble Codes" are information about faults when a recording trigger is established. Various diagnostic trouble codes could be set and recorded due to component or system damage during an accident.
- The airbag ECU records only diagnostic information related to the airbag system. It does not record diagnostic information related to other vehicle systems.
- The TaSCAN, Global TechStream, or Intelligent Tester II devices (or any other Toyota genuine diagnostic tool) can be used to obtain detailed information on the diagnostic trouble codes from the airbag system, as well as diagnostic information from other systems. However, in some cases, the diagnostic trouble codes of the airbag system recorded by the airbag ECU when the event occurred may not match the diagnostic trouble codes read out when the diagnostic tool is used.

General Information:

- The data recording specifications of Toyota's airbag ECUs are divided into the following seven categories. The specifications for 12EDR or later are designed to be compatible with NHTSA's 49CFR Part 563 rule.
  - 00EDR / 02EDR / 04EDR / 06EDR / 10EDR / 12EDR / 13EDR
- The airbag ECU records data for all or some of the following accident types: frontal crash, rear crash , side crash, and rollover events. Depending on the installed airbag ECU, data for side crash and/or rollover events may not be recorded.
- This airbag ECU records post-crash data, and depending on the airbag ECU, may record pre-crash data.
  - If a single event occurs independently, the data for that event is recorded on a one-to-one basis.
  - If multiple events occur successively (within a period of approximately 500ms), the establishment of the recording trigger for the first event is defined as the "pre-crash recording trigger". Pre-crash data for the first event and post-crash data for each successive event is then recorded.
  - In some situations, pre-crash data may be re-recorded if an event continues for more than 0.5 seconds.
- The airbag ECU has two recording pages (memory maps) to store pre-crash data. Additionally, to store post-crash data, the airbag ECU has two recording pages for each accident type: two pages for frontal and rear crash, two pages for a side crash, and two pages for rollover event.
- The data recorded by the airbag ECU includes correlating information between each previously occurring event (i.e., information that clarifies the collision event sequence. This correlation information consists of the following items.
  - Time from Previous Pre-Crash TRG
  - Linked Pre-Crash Page

- Time from Pre-Crash TRG
- TRG Count
- Previous Crash Type

- The point in time at which the recording trigger is established is regarded as time zero for the recorded data.
- The recording trigger judgment threshold value differs depending on the collision type (i.e., frontal crash, rear crash, side crash, or rollover event).
- Time series data for side crash may have 24 or 25 sampling points.
- Some of the data recorded by the airbag ECU is transmitted to the airbag ECU from various vehicle control modules by the vehicle's Controller Area Network (CAN).
- In some cases, the airbag ECU part number printed on the ECU label may not match the airbag ECU part number that the CDR tool reports. The part number retrieved by the CDR tool should be considered as the official ECU part number.

Data Element Sign Convention:
The following table provides an explanation of the sign notation for data elements that may be included in this CDR report.

Data Element Name	Positive Sign Notation Indicates
Max. Longitudinal Delta-V	Forward
Longitudinal Delta-V	Forward
Max. Lateral Delta-V , B-Pillar Sensor	Outside to Inside
Max. Lateral Delta-V , C-Pillar Sensor	Outside to Inside
Max. Lateral Delta-V , Front Door Sensor	Outside to Inside
Max. Lateral Delta-V , Slide Door Sensor	Outside to Inside
Lateral Delta-V , B-Pillar Sensor	Outside to Inside
Lateral Delta-V , C-Pillar Sensor	Outside to Inside
Lateral Delta-V , Airbag ECU Sensor	Outside to Inside
Roll Angle Peak	Clockwise Rotation
Roll Angle	Clockwise Rotation
Lateral Acceleration , Airbag ECU Sensor *	Right to Left

* For sensing a rollover

Data Definitions:
1)

- The "ON" setting for the "Freeze Signal" indicates a state in which the non-volatile memory can not be overwritten or deleted by the airbag ECU. After "Freeze Signal" has been turned ON, subsequent events will not be recorded.
- "Recording Status" indicates a state in which all recorded event data has been written into the non-volatile memory, or a state in which this process was interrupted and not fully written into the non-volatile memory. If "Recording Status" is "Incomplete", recorded event data may not be valid.
- "Time to Deployment Command" indicates the time between recording trigger establishment and the determination of airbag deployment. This value may differ from the actual time it takes for the airbag to fully deploy.
- Even if an airbag/pretensioner did not deploy due to the "front passenger airbag disable switch and/or "RSCA Disable Switch" in the ON position or other disabling criteria are met, the "Time to deployment command" data element for that airbag/pretensioner may still be recorded.
- "Engine RPM" indicates the number of engine revolutions, not the number of motor revolutions. The recorded value has an upper limit of 5,200 rpm. Resolution is 400 rpm and the value is rounded down and recorded. For example, if the actual engine speed is 799 rpm, the recorded value will be 400 rpm.
- The upper limit for the recorded "Vehicle Speed" value is 122 km/h (75.8mph). Resolution is 2km/h (1.2mph) and the value is rounded down and recorded. The accuracy of the "Vehicle Speed" value can be affected by various factors. These include, but not limited, to the following.
  - Significant changes in the tire's rolling radius
  - Wheel lock and wheel slip
- "Accelerator Rate" has two recording specifications. Both the recorded value increases as the driver depresses the accelerator.
  - Percentage of accelerator pedal depressed (recorded as 0-100(%)).
  - Output voltage of accelerator pedal module (recorded as 0-5(V)).
- The "Drive" setting for the "Shift Position" value indicates the shift position state is other than "R,"(Reverse), "N" (Neutral), or "P" (Park). If sequential shift had been used, "Invalid" may be displayed.
- Depending on the type of occupant sensor installed in the vehicle, one of the following four recording formats for "Occupancy Status, Passenger" will be utilized.
  - Occupied / Not Occupied
  - Adult / Child / Not Occupied
  - AM50 / AF05 / Child / Not Occupied
  - AM50 / AF05 / Child or Not Occupied
- "Ignition Cycle Since DTC was Set" records the cumulative amount of times that the ignition is switched ON after an airbag system diagnostics code was set for the first time.
- "Air Bag Warning Lamp ON Time Since DTC was Set" records the total time that the ignition has been switched ON after the warning lamp was illuminate, due to an airbag system fault, for the first time. The resolution is 15 minutes, and the value is rounded down and recorded.
- "Longitudinal Delta-V" indicates the change in forward speed after establishment of the recording trigger. This does not refer to vehicle speed, and it does not include the change in speed during the period from the start of the actual collision to establishment of the recording trigger.

- "Roll Angle peak" may not always match the peak value within the "Roll Angle" sampling points due to differences in data calculation method.
- For "Lateral Delta-V", the sensor location (B-pillar, front door, C-pillar, and slide door) shows the outline of a typical sensor position. Sensory location can be confirmed using the repair manual.
- "Time from Previous Pre-Crash TRG" indicates the time between the establishment of an event's pre-crash recording trigger to the establishment of a more recent event's pre-crash recording trigger. The upper limit for the recorded value is 16,381 milliseconds. In the event of establishment of the first pre-crash recording trigger after the ignition is switched ON, the upper limit value(max value) is recorded.
- "TRG Count" indicates a calculated value of the number of times recording triggers have been established for all crash types. The sequence in which each event occurred can be verified from the "TRG Count". The smaller the "TRG Count" value, the older the data. The upper limit for the recorded value is 65,533 times. When more than one event reaches the upper limit, the actual "TRG Count" may be greater than what is displayed for that event.
- "Linked Pre-Crash Page" is used to link 'paged" pre-crash data with 'paged" post-crash data. When old pre-crash data is overwritten by new pre-crash data, the "Linked Pre-Crash Page" value may record a page number that is not actually linked.
- Resolution of the "Time from Pre-Crash to TRG" is 100 [ms], and the value is rounded down and recorded.

05006_ToyotaF-TEN_r023

System Status at Time of Retrieval

ECU Part Number	89170-52D80
ECU Generation	06EDR
Recording Status, All Pages	Complete
Freeze Signal	ON
Freeze Signal Factor	Front Airbag Deployment Front Pretensioner Deployment
Diagnostic Trouble Codes Exist	No
Time from Previous Pre Crash TRG (msec)	564
Latest Pre-Crash Page	0
Contains Unlinked Pre-Crash Data	Yes

Event Record Summary at Retrieval

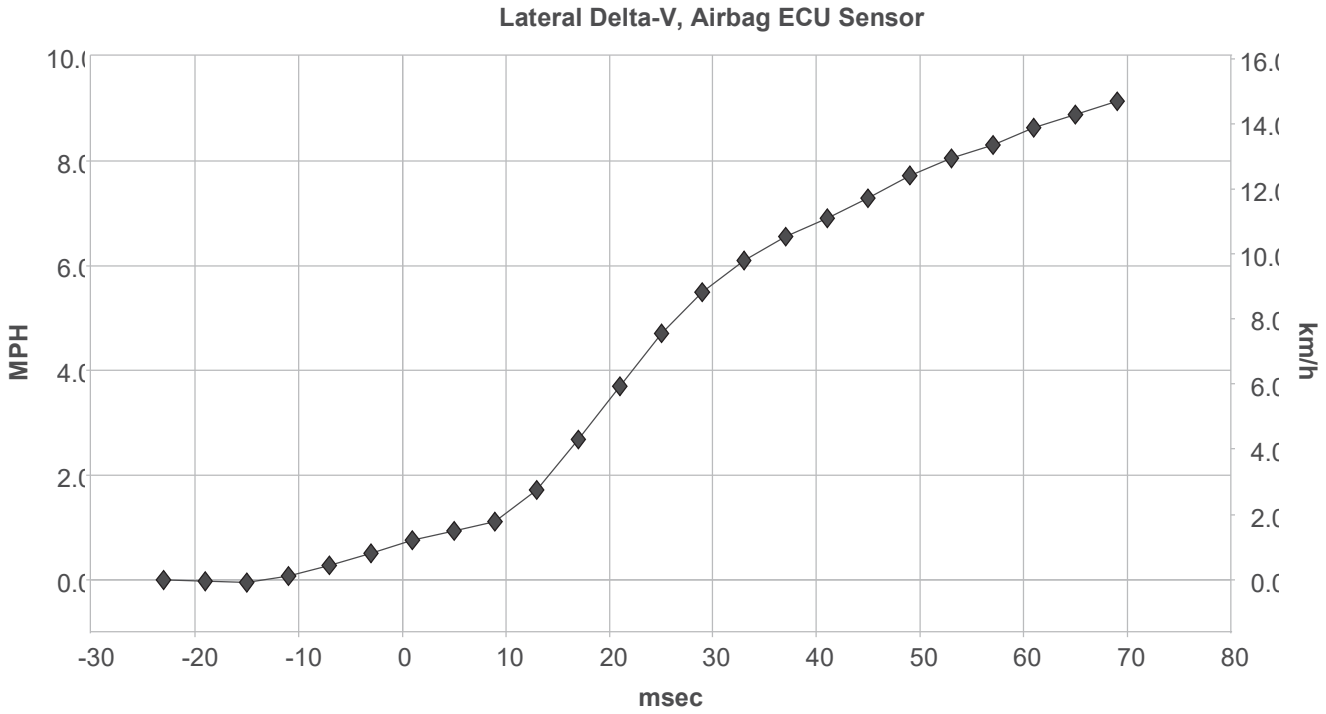
Events Recorded	TRG Count	Crash Type	Time (msec)	Pre-Crash & DTC Data Recording Status	Event & Crash Pulse Data Recording Status
Most Recent Event	7	Side Crash	0	Complete (Page 0)	Complete (Side Page 0)
1st Prior Event	6	Front/Rear Crash	-20	Complete (Page 0)	Complete (Front/Rear Page 1)
2nd Prior Event	5	Side Crash	-228	Complete (Page 0)	Complete (Side Page 1)
3rd Prior Event	4	Front/Rear Crash	-282	Complete (Page 0)	Complete (Front/Rear Page 0)

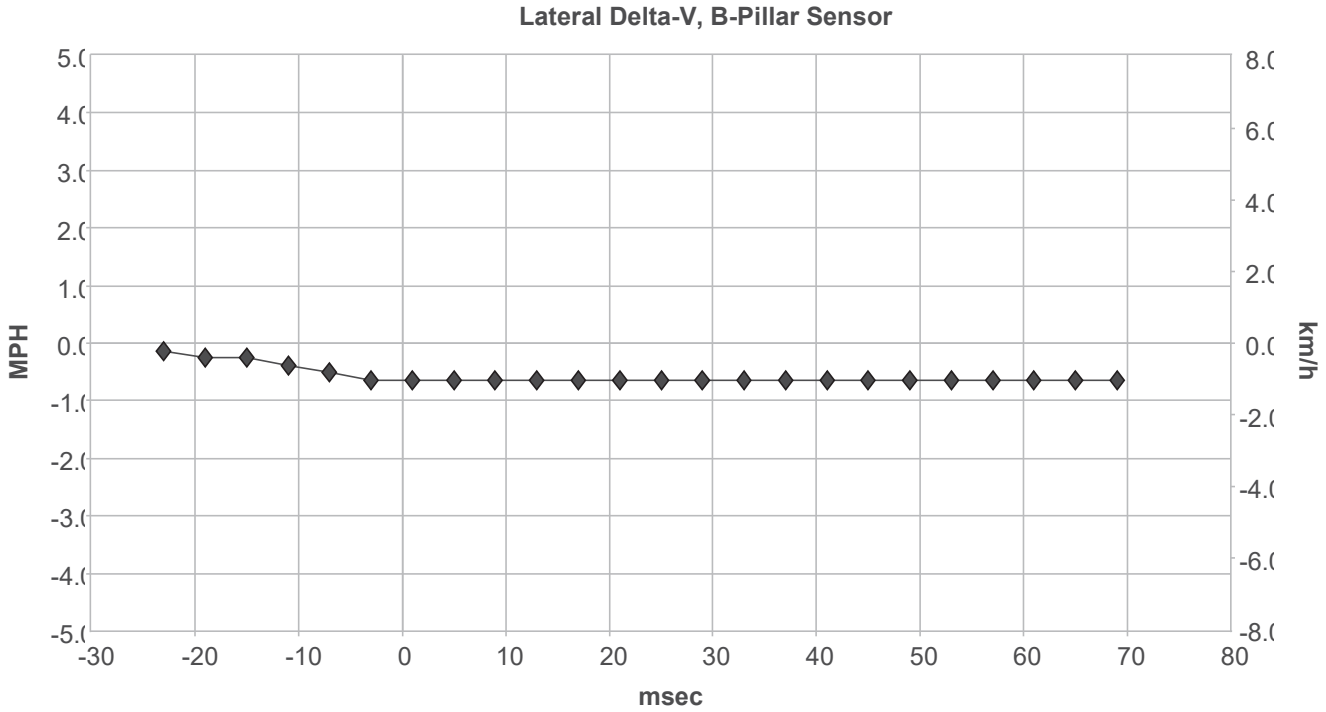
System Status at Event (Most Recent Event, TRG 7)

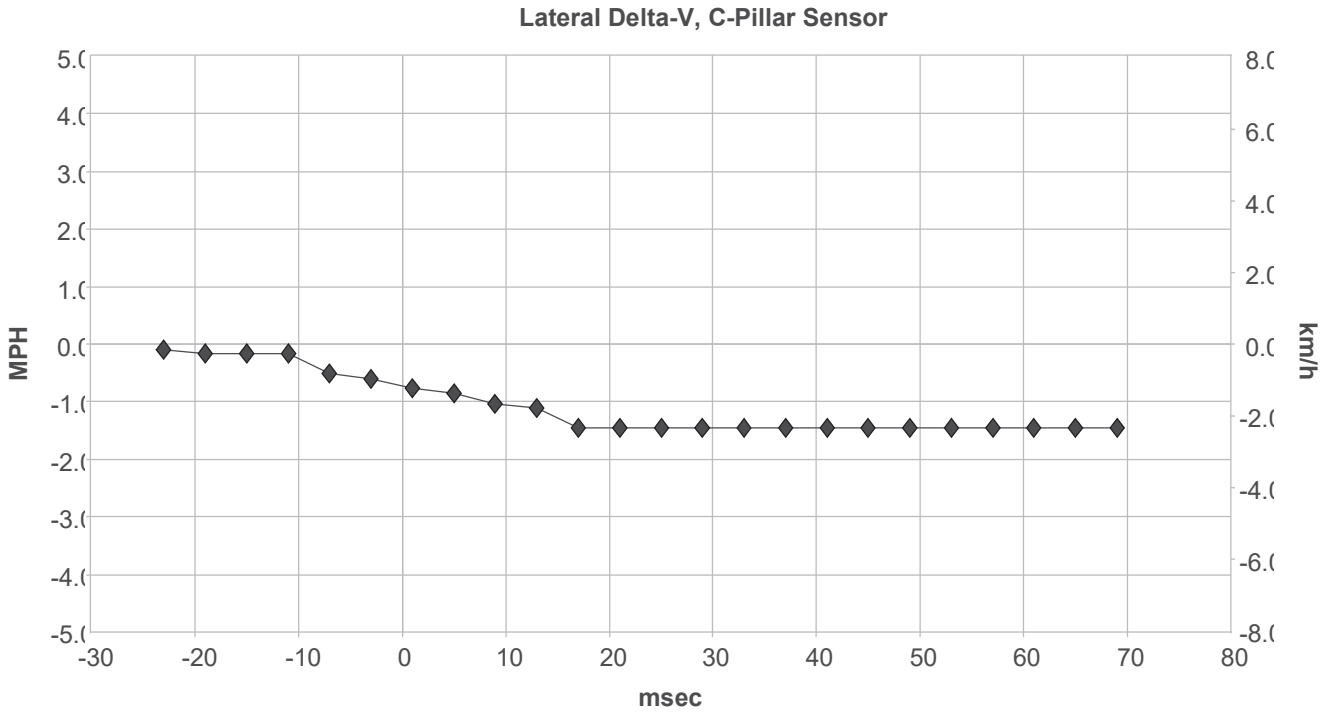
Recording Status, Side Crash Info.	Complete
Crash Type	Side Crash
TRG Count (times)	7
Recorded Side	Passenger's Side
Previous Crash Type	Frontal/Rear
Time from Pre-Crash TRG (msec)	282
Linked Pre-Crash Page	0
Time to Deployment Command, B-Pillar Sensor (msec)	Not Commanded
Time to Deployment Command, C-Pillar Sensor (msec)	Not Commanded

Lateral Crash Pulse (Most Recent Event, TRG 7 - table 1 of 2)

Recording Status, Time Series Data	Complete
Time from TRG to Next Sample (msec)	1
Max Lateral Delta-V, B-Pillar Sensor (MPH [km/h])	-0.6 [-1.0]
Max Lateral Delta-V, C-Pillar Sensor (MPH [km/h])	-1.5 [-2.3]






Lateral Delta-V, C-Pillar Sensor
MPH
km/h
msec
5.0
4.0
3.0
2.0
1.0
0.0
-1.0
-2.0
-3.0
-4.0
-5.0
8.0
6.0
4.0
2.0
0.0
-2.0
-4.0
-6.0
-8.0
-30
-20
-10
0
10
20
30
40
50
60
70
80

Lateral Crash Pulse (Most Recent Event, TRG 7 - table 2 of 2)

Time (msec)	Lateral Delta-V, Airbag ECU Sensor (MPH [km/h])	Lateral Delta-V, B-Pillar Sensor (MPH [km/h])	Lateral Delta-V, C-Pillar Sensor (MPH [km/h])
-23	0.0 [0.0]	-0.1 [-0.2]	-0.1 [-0.1]
-19	0.0 [-0.1]	-0.3 [-0.4]	-0.2 [-0.3]
-15	-0.1 [-0.1]	-0.3 [-0.4]	-0.2 [-0.3]
-11	0.1 [0.1]	-0.4 [-0.6]	-0.2 [-0.3]
-7	0.3 [0.4]	-0.5 [-0.8]	-0.5 [-0.8]
-3	0.5 [0.8]	-0.6 [-1.0]	-0.6 [-1.0]
1	0.7 [1.2]	-0.6 [-1.0]	-0.8 [-1.2]
5	0.9 [1.5]	-0.6 [-1.0]	-0.9 [-1.4]
9	1.1 [1.8]	-0.6 [-1.0]	-1.0 [-1.7]
13	1.7 [2.8]	-0.6 [-1.0]	-1.1 [-1.8]
17	2.7 [4.3]	-0.6 [-1.0]	-1.5 [-2.3]
21	3.7 [5.9]	-0.6 [-1.0]	-1.5 [-2.3]
25	4.7 [7.6]	-0.6 [-1.0]	-1.5 [-2.3]
29	5.5 [8.8]	-0.6 [-1.0]	-1.5 [-2.3]
33	6.1 [9.8]	-0.6 [-1.0]	-1.5 [-2.3]
37	6.6 [10.6]	-0.6 [-1.0]	-1.5 [-2.3]
41	6.9 [11.1]	-0.6 [-1.0]	-1.5 [-2.3]
45	7.3 [11.7]	-0.6 [-1.0]	-1.5 [-2.3]
49	7.7 [12.4]	-0.6 [-1.0]	-1.5 [-2.3]
53	8.1 [13.0]	-0.6 [-1.0]	-1.5 [-2.3]
57	8.3 [13.4]	-0.6 [-1.0]	-1.5 [-2.3]
61	8.6 [13.9]	-0.6 [-1.0]	-1.5 [-2.3]
65	8.9 [14.3]	-0.6 [-1.0]	-1.5 [-2.3]
69	9.1 [14.7]	-0.6 [-1.0]	-1.5 [-2.3]

DTCs Present at Time of Event (Most Recent Event, TRG 7)

Recording Status, Diagnostic	Complete
Ignition Cycle Since DTC was Set (times)	0
Airbag Warning Lamp ON Time Since DTC was Set (min)	0
Diagnostic Trouble Codes	None

Pre-Crash Data, 1 Sample (Most Recent Event, TRG 7)

Recording Status, Pre-Crash/Occupant	Complete
Time from Pre-Crash to TRG (msec)	0
Buckle Switch, Driver	Buckled
Buckle Switch, Passenger	Unbuckled
Occupancy Status, Passenger	Not Occupied
Seat Position, Driver	Rearward
Shift Position	Drive

Pre-Crash Data, -5 to 0 seconds (Most Recent Event, TRG 7)

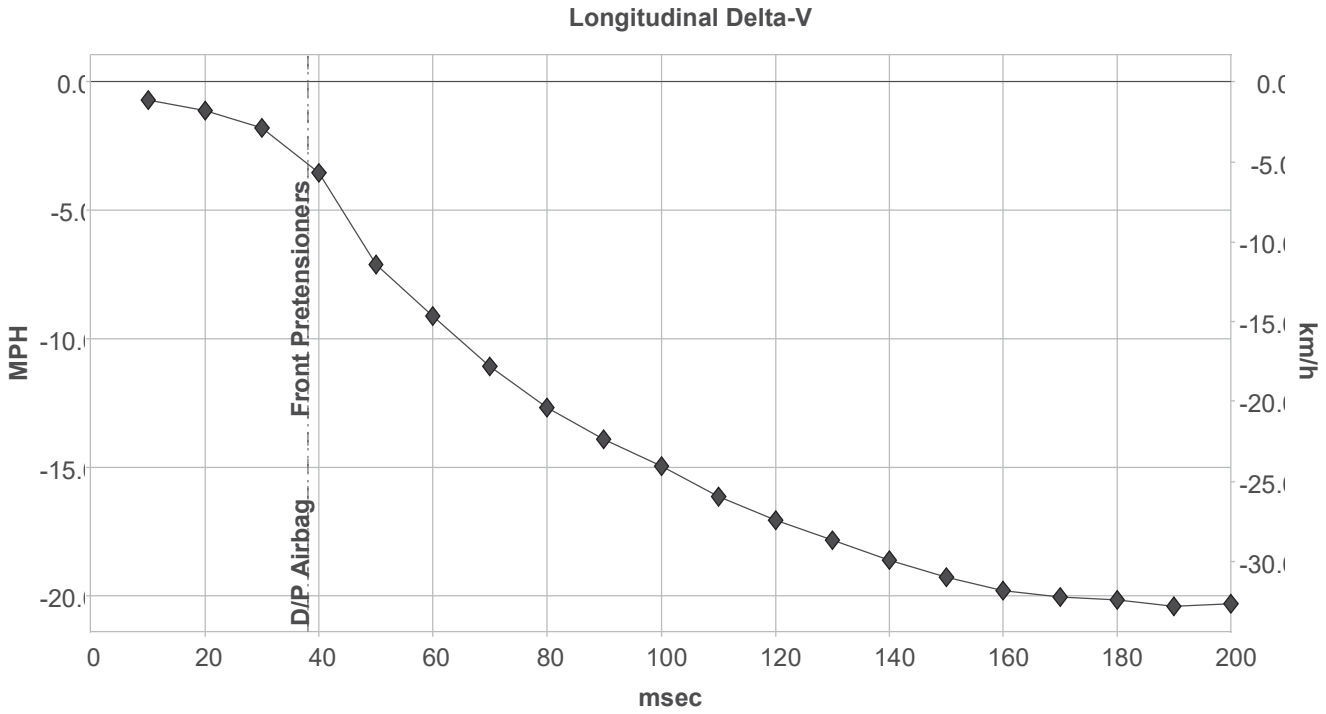
Time (sec)	-4	-3	-2	-1	0	0 (TRG)
Vehicle Speed (MPH [km/h])	75.8 [122]	73.3 [118]	48.5 [78]	42.3 [68]	22.4 [36]	22.4 [36]
Brake Switch	ON	ON	ON	ON	ON	ON
Accelerator Rate (V)	0.78	0.78	0.78	0.78	0.78	0.78
Engine RPM (RPM)	3,200	2,800	2,000	1,600	800	800

System Status at Event (1st Prior Event, TRG 6)

Recording Status, Front/Rear Crash Info.	Complete
Crash Type	Front/Rear Crash
TRG Count (times)	6
Previous Crash Type	Side
Time from Pre-Crash TRG (msec)	262
Linked Pre-Crash Page	0
Time to Deployment Command, Front Airbag, Driver (msec)	38
Time to Deployment Command, Front Airbag, Passenger (msec)	38
Event Severity Status, Driver	Level 3
Event Severity Status, Passenger	N/A
Time to Deployment Command, Pretensioner (msec)	38

Longitudinal Crash Pulse (1st Prior Event, TRG 6 - table 1 of 2)

Recording Status, Time Series Data	Complete
Max Longitudinal Delta-V (MPH [km/h])	-20.4 [-32.8]



Longitudinal Crash Pulse (1st Prior Event, TRG 6 - table 2 of 2)

Time (msec)	Longitudinal Delta-V (MPH [km/h])
10	-0.7 [-1.1]
20	-1.1 [-1.8]
30	-1.8 [-2.9]
40	-3.5 [-5.7]
50	-7.1 [-11.4]
60	-9.1 [-14.6]
70	-11.1 [-17.8]
80	-12.7 [-20.4]
90	-13.9 [-22.3]
100	-14.9 [-24.0]
110	-16.1 [-25.9]
120	-17.1 [-27.4]
130	-17.8 [-28.7]
140	-18.6 [-29.9]
150	-19.3 [-31.0]
160	-19.8 [-31.9]
170	-20.1 [-32.3]
180	-20.1 [-32.4]
190	-20.4 [-32.8]
200	-20.3 [-32.7]

DTCs Present at Time of Event (1st Prior Event, TRG 6)

Recording Status, Diagnostic	Complete
Ignition Cycle Since DTC was Set (times)	0
Airbag Warning Lamp ON Time Since DTC was Set (min)	0
Diagnostic Trouble Codes	None

Pre-Crash Data, 1 Sample (1st Prior Event, TRG 6)

Recording Status, Pre-Crash/Occupant	Complete
Time from Pre-Crash to TRG (msec)	0
Buckle Switch, Driver	Buckled
Buckle Switch, Passenger	Unbuckled
Occupancy Status, Passenger	Not Occupied
Seat Position, Driver	Rearward
Shift Position	Drive

Pre-Crash Data, -5 to 0 seconds (1st Prior Event, TRG 6)

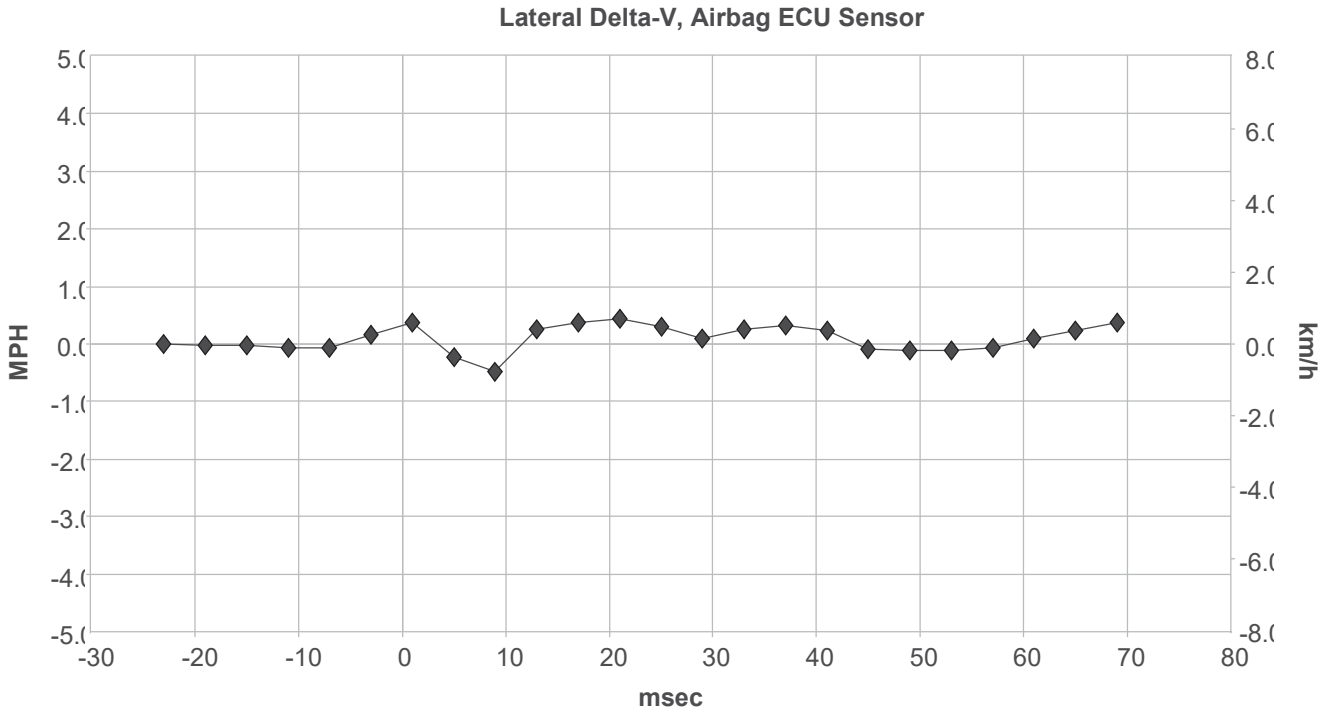
Time (sec)	-4	-3	-2	-1	0	0 (TRG)
Vehicle Speed (MPH [km/h])	75.8 [122]	73.3 [118]	48.5 [78]	42.3 [68]	22.4 [36]	22.4 [36]
Brake Switch	ON	ON	ON	ON	ON	ON
Accelerator Rate (V)	0.78	0.78	0.78	0.78	0.78	0.78
Engine RPM (RPM)	3,200	2,800	2,000	1,600	800	800

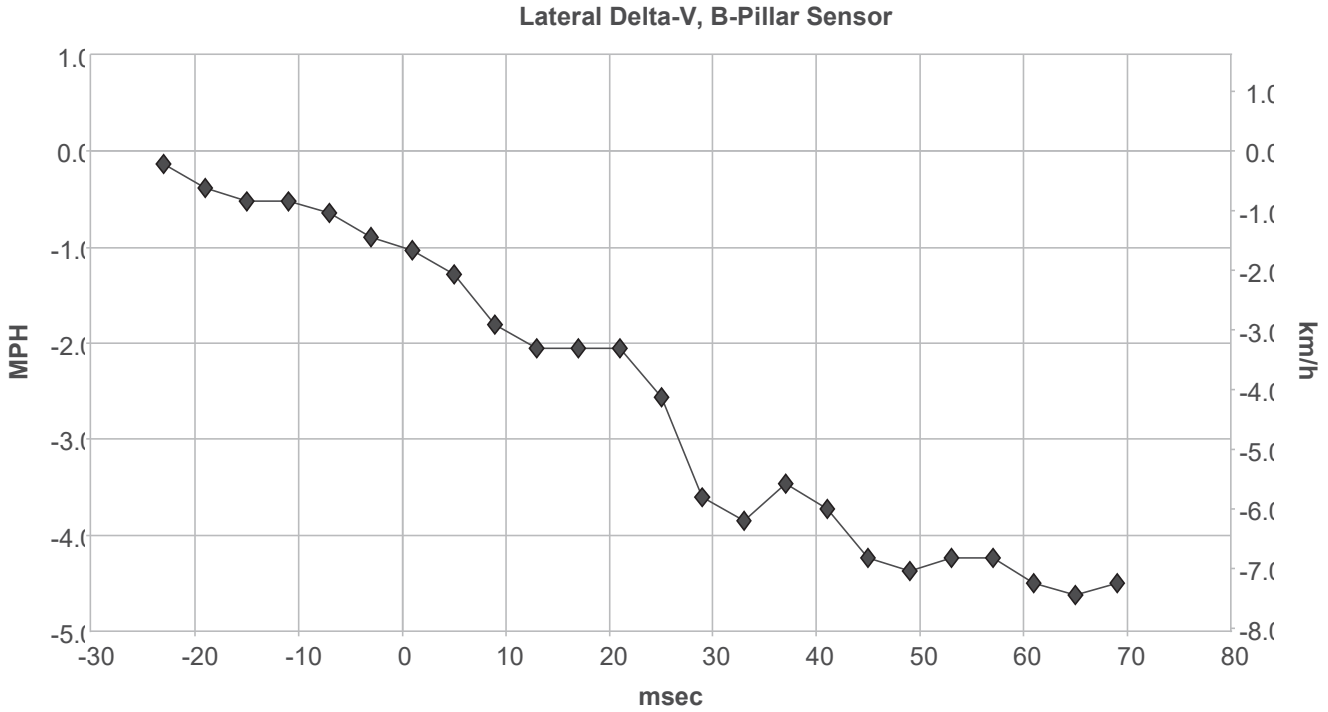
System Status at Event (2nd Prior Event, TRG 5)

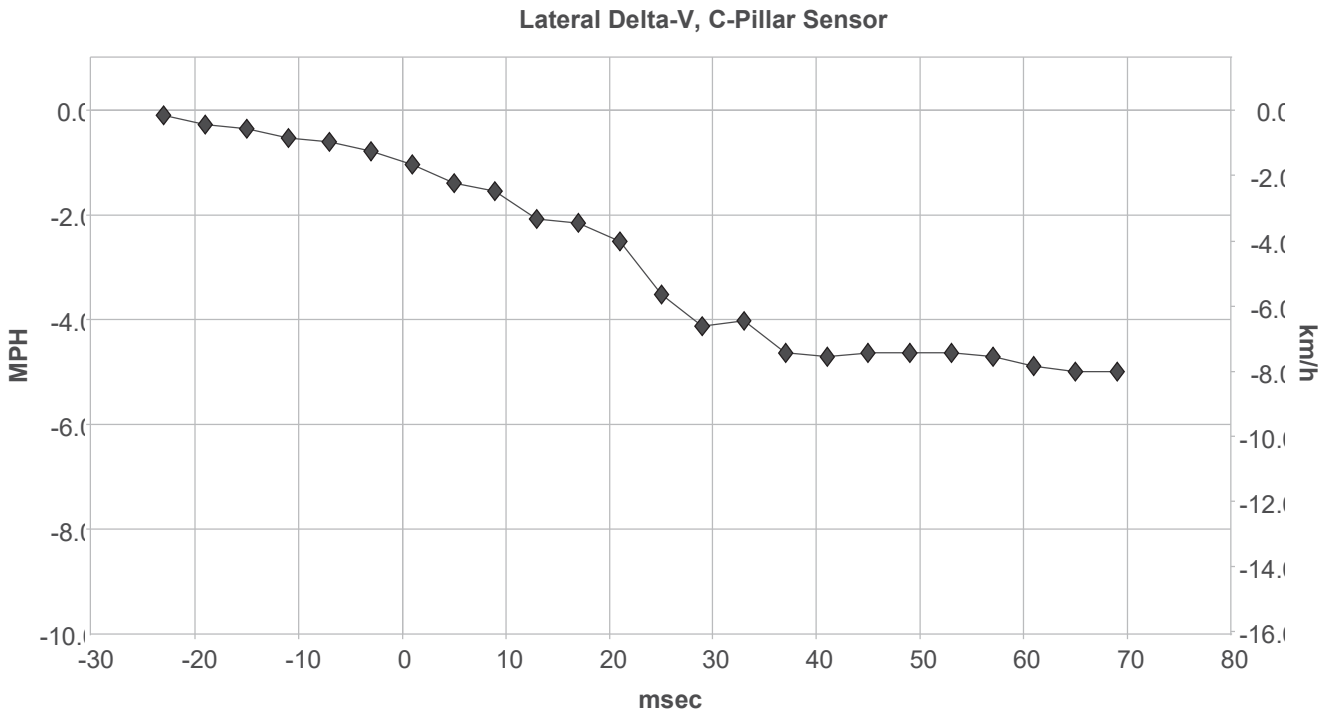
Recording Status, Side Crash Info.	Complete
Crash Type	Side Crash
TRG Count (times)	5
Recorded Side	Passenger's Side
Previous Crash Type	Frontal/Rear
Time from Pre-Crash TRG (msec)	54
Linked Pre-Crash Page	0
Time to Deployment Command, B-Pillar Sensor (msec)	Not Commanded
Time to Deployment Command, C-Pillar Sensor (msec)	Not Commanded

Lateral Crash Pulse (2nd Prior Event, TRG 5 - table 1 of 2)

Recording Status, Time Series Data	Complete
Time from TRG to Next Sample (msec)	1
Max Lateral Delta-V, B-Pillar Sensor (MPH [km/h])	-4.6 [-7.4]
Max Lateral Delta-V, C-Pillar Sensor (MPH [km/h])	-5.0 [-8.0]






Lateral Delta-V, C-Pillar Sensor
MPH
km/h
msec
0.0
-2.0
-4.0
-6.0
-8.0
-10.0
-30
-20
-10
0
10
20
30
40
50
60
70
80

Lateral Crash Pulse (2nd Prior Event, TRG 5 - table 2 of 2)

Time (msec)	Lateral Delta-V, Airbag ECU Sensor (MPH [km/h])	Lateral Delta-V, B-Pillar Sensor (MPH [km/h])	Lateral Delta-V, C-Pillar Sensor (MPH [km/h])
-23	0.0 [0.0]	-0.1 [-0.2]	-0.1 [-0.1]
-19	0.0 [0.0]	-0.4 [-0.6]	-0.3 [-0.4]
-15	0.0 [0.0]	-0.5 [-0.8]	-0.3 [-0.6]
-11	-0.1 [-0.1]	-0.5 [-0.8]	-0.5 [-0.8]
-7	-0.1 [-0.1]	-0.6 [-1.0]	-0.6 [-1.0]
-3	0.2 [0.2]	-0.9 [-1.4]	-0.8 [-1.2]
1	0.4 [0.6]	-1.0 [-1.7]	-1.0 [-1.7]
5	-0.2 [-0.4]	-1.3 [-2.1]	-1.4 [-2.2]
9	-0.5 [-0.8]	-1.8 [-2.9]	-1.5 [-2.5]
13	0.3 [0.4]	-2.1 [-3.3]	-2.1 [-3.3]
17	0.4 [0.6]	-2.1 [-3.3]	-2.1 [-3.4]
21	0.4 [0.7]	-2.1 [-3.3]	-2.5 [-4.0]
25	0.3 [0.5]	-2.6 [-4.1]	-3.5 [-5.7]
29	0.1 [0.1]	-3.6 [-5.8]	-4.1 [-6.6]
33	0.3 [0.4]	-3.9 [-6.2]	-4.0 [-6.5]
37	0.3 [0.5]	-3.5 [-5.6]	-4.6 [-7.4]
41	0.2 [0.4]	-3.7 [-6.0]	-4.7 [-7.6]
45	-0.1 [-0.2]	-4.2 [-6.8]	-4.6 [-7.4]
49	-0.1 [-0.2]	-4.4 [-7.0]	-4.6 [-7.4]
53	-0.1 [-0.2]	-4.2 [-6.8]	-4.6 [-7.4]
57	-0.1 [-0.1]	-4.2 [-6.8]	-4.7 [-7.6]
61	0.1 [0.1]	-4.5 [-7.2]	-4.9 [-7.9]
65	0.2 [0.4]	-4.6 [-7.4]	-5.0 [-8.0]
69	0.4 [0.6]	-4.5 [-7.2]	-5.0 [-8.0]

DTCs Present at Time of Event (2nd Prior Event, TRG 5)

Recording Status, Diagnostic	Complete
Ignition Cycle Since DTC was Set (times)	0
Airbag Warning Lamp ON Time Since DTC was Set (min)	0
Diagnostic Trouble Codes	None

Pre-Crash Data, 1 Sample (2nd Prior Event, TRG 5)

Recording Status, Pre-Crash/Occupant	Complete
Time from Pre-Crash to TRG (msec)	0
Buckle Switch, Driver	Buckled
Buckle Switch, Passenger	Unbuckled
Occupancy Status, Passenger	Not Occupied
Seat Position, Driver	Rearward
Shift Position	Drive

Pre-Crash Data, -5 to 0 seconds (2nd Prior Event, TRG 5)

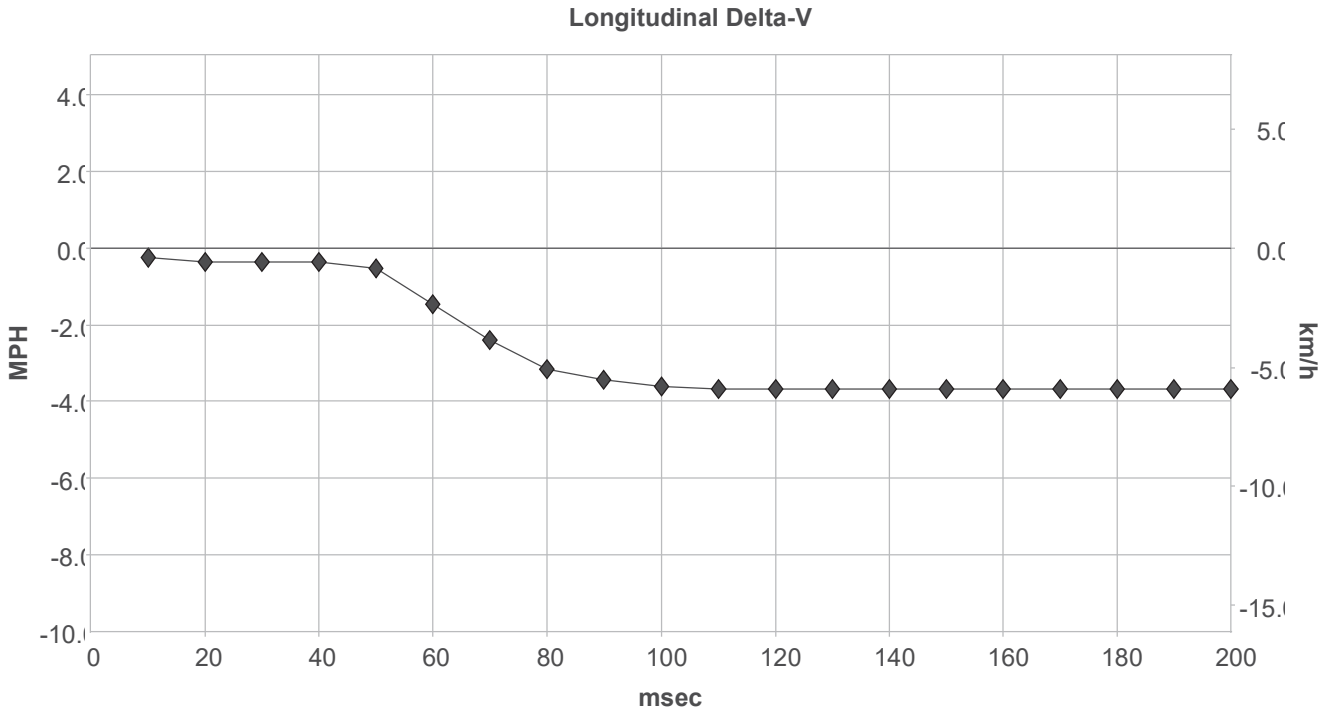
Time (sec)	-4	-3	-2	-1	0	0 (TRG)
Vehicle Speed (MPH [km/h])	75.8 [122]	73.3 [118]	48.5 [78]	42.3 [68]	22.4 [36]	22.4 [36]
Brake Switch	ON	ON	ON	ON	ON	ON
Accelerator Rate (V)	0.78	0.78	0.78	0.78	0.78	0.78
Engine RPM (RPM)	3,200	2,800	2,000	1,600	800	800

System Status at Event (3rd Prior Event, TRG 4)

Recording Status, Front/Rear Crash Info.	Complete
Crash Type	Front/Rear Crash
TRG Count (times)	4
Previous Crash Type	Side
Time from Pre-Crash TRG (msec)	0
Linked Pre-Crash Page	0
Time to Deployment Command, Front Airbag, Driver (msec)	Not Commanded
Time to Deployment Command, Front Airbag, Passenger (msec)	Not Commanded
Event Severity Status, Driver	N/A
Event Severity Status, Passenger	N/A
Time to Deployment Command, Pretensioner (msec)	Not Commanded

Longitudinal Crash Pulse (3rd Prior Event, TRG 4 - table 1 of 2)

Recording Status, Time Series Data	Complete
Max Longitudinal Delta-V (MPH [km/h])	-3.7 [-5.9]



Longitudinal Crash Pulse (3rd Prior Event, TRG 4 - table 2 of 2)

Time (msec)	Longitudinal Delta-V (MPH [km/h])
10	-0.3 [-0.4]
20	-0.3 [-0.6]
30	-0.3 [-0.6]
40	-0.3 [-0.6]
50	-0.5 [-0.8]
60	-1.5 [-2.3]
70	-2.4 [-3.9]
80	-3.2 [-5.1]
90	-3.4 [-5.5]
100	-3.6 [-5.8]
110	-3.7 [-5.9]
120	-3.7 [-5.9]
130	-3.7 [-5.9]
140	-3.7 [-5.9]
150	-3.7 [-5.9]
160	-3.7 [-5.9]
170	-3.7 [-5.9]
180	-3.7 [-5.9]
190	-3.7 [-5.9]
200	-3.7 [-5.9]

DTCs Present at Time of Event (3rd Prior Event, TRG 4)

Recording Status, Diagnostic	Complete
Ignition Cycle Since DTC was Set (times)	0
Airbag Warning Lamp ON Time Since DTC was Set (min)	0
Diagnostic Trouble Codes	None

Pre-Crash Data, 1 Sample (3rd Prior Event, TRG 4)

Recording Status, Pre-Crash/Occupant	Complete
Time from Pre-Crash to TRG (msec)	0
Buckle Switch, Driver	Buckled
Buckle Switch, Passenger	Unbuckled
Occupancy Status, Passenger	Not Occupied
Seat Position, Driver	Rearward
Shift Position	Drive

Pre-Crash Data, -5 to 0 seconds (3rd Prior Event, TRG 4)

Time (sec)	-4	-3	-2	-1	0	0 (TRG)
Vehicle Speed (MPH [km/h])	75.8 [122]	73.3 [118]	48.5 [78]	42.3 [68]	22.4 [36]	22.4 [36]
Brake Switch	ON	ON	ON	ON	ON	ON
Accelerator Rate (V)	0.78	0.78	0.78	0.78	0.78	0.78
Engine RPM (RPM)	3,200	2,800	2,000	1,600	800	800

DTCs Present at Time of Event (Unlinked, Page 1)

Recording Status, Diagnostic	Complete
Ignition Cycle Since DTC was Set (times)	0
Airbag Warning Lamp ON Time Since DTC was Set (min)	0
Diagnostic Trouble Codes	None

Pre-Crash Data, 1 Sample (Unlinked, Page 1)

Recording Status, Pre-Crash/Occupant	Complete
Time from Pre-Crash to TRG (msec)	500
Buckle Switch, Driver	Buckled
Buckle Switch, Passenger	Unbuckled
Occupancy Status, Passenger	Not Occupied
Seat Position, Driver	Rearward
Shift Position	Drive

Pre-Crash Data, -5 to 0 seconds (Unlinked, Page 1)

Time (sec)	-4.5	-3.5	-2.5	-1.5	-0.5	0 (TRG)
Vehicle Speed (MPH [km/h])	75.8 [122]	75.8 [122]	73.3 [118]	48.5 [78]	42.3 [68]	36 [58]
Brake Switch	ON	ON	ON	ON	ON	ON
Accelerator Rate (V)	0.78	0.78	0.78	0.78	0.78	0.78
Engine RPM (RPM)	3,600	3,200	2,800	2,000	1,600	1,200